llmXive Automated Review of Why Can't I Open My Drawer? Mitigating Object-Driven Shortcuts in Zero-Shot Compositional Action Recognition

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a coherent diagnosis of object-driven shortcuts in ZS-CAR and proposes a method to mitigate them. The diagnostic metrics are well-defined and the experimental setup is rigorous. However, there are two specific instances where the text makes claims that do not align with the cited evidence or tables.

First, in Section 3.2, the authors state that the baseline C2C yields a cosine similarity of 0.92 between forward and reversed verb features, citing Figure 3. However, Figure 3 (fig_diagnosis_baseline) displays learning curves for FSP/FCP and accuracy, not cosine similarity. The 0.92 value is actually presented in the supplementary Figure S1 (fig_cosine_sim). This is a citation mismatch that should be corrected to point to the correct figure.

Second, in Section 5.2, the text claims that "Jung et al. yields the lowest unseen composition accuracy among the CLIP baselines" on Sth-com. According to Table 1 (sth_com.tex), the AIM baseline achieves 39.19% unseen accuracy, while Jung et al. achieves 53.55%. Since AIM is lower, the claim is factually incorrect as stated. The authors should clarify that Jung et al. has the lowest accuracy among the *compositional* baselines (excluding simple adapter baselines like AIM) or correct the statement to reflect the actual data.

These are minor issues that can be resolved with text edits and do not affect the core validity of the proposed method or the main experimental conclusions.

Action items.

- [**writing**] Section 3.2 cites Fig 3 for a cosine similarity of 0.92, but Fig 3 shows FSP/FCP curves. The value is in supplementary Fig S1. Update the citation to the correct figure or move the analysis to the main text.
- [**writing**] Section 5.2 claims Jung et al. has the lowest unseen accuracy among CLIP baselines, but Table 1 shows AIM (39.19%) is lower than Jung (53.55%). Qualify the claim to 'lowest among compositional baselines' or correct the comparison set.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure is visually clear but scientifically opaque due to a completely missing caption and undefined acronyms in the legend, making it impossible to interpret the data's significance.

Figure 2

The figure is visually cluttered due to an internal legend and lacks a descriptive caption, which hinders understanding of the plotted metrics and their significance.

Figure 3

The figure is visually clear with well-labeled axes and data points, but it lacks a descriptive caption to explain the specific experimental setup or the distinction between the two 'False' prediction metrics.

Figure 4

The figure contains clear visualizations of learning curves and heatmaps, but the complete absence of a descriptive caption prevents verification of the data and legends shown.

Figure 5

The figure consists of a visual sequence labeled with an action, but the complete absence of a caption makes it impossible to assess its scientific relevance or verify its content against the paper's claims.

Figure 6

The bar chart clearly displays the top 30 frequent compositions, but the x-axis labels are crowded and the figure lacks a descriptive caption to explain its significance.

Figure 7

The figure is visually clear with labeled axes and data points, but it lacks a descriptive caption and relies on undefined acronyms in the legend, hindering standalone interpretation.

Figure 8

Figure 8 provides a clear visual overview of the model architecture and regularization methods, but the complete absence of a caption makes the specific variables, loss functions, and panel relationships difficult to interpret for the reader.

Figure 9

Figure 9 effectively visualizes the experimental setup using a grid format, but it is critically hindered by the absence of a descriptive caption and an incomplete legend for the first panel, leaving the specific meaning of the empty cells undefined.

Figure 10

Figure 10 is a clear and well-structured flowchart that effectively illustrates the evaluation logic for distinguishing between seen and unseen compositions. The visual hierarchy, decision nodes, and resulting metrics are unambiguous and self-explanatory without requiring a detailed caption.

Figure 11

Figure 11 presents two heatmaps illustrating a dataset split for training and testing, but the complete absence of a caption makes the specific definitions of the 'Seen' and 'Unseen' categories and the experimental goal impossible to verify.

Figure 12

Figure 12 is a clear, well-structured schematic that effectively communicates the paper's core problem: object-driven shortcuts in zero-shot action recognition. The three panels logically progress from data priors to learning difficulty to the specific failure mode in testing, with all visual elements (icons, arrows, text) being legible and self-explanatory.

Action items.

- [**fatal**] Figure 1: The caption is explicitly '(no caption)', providing no context for the plot's axes, data series, or scientific claim.
- [**science**] Figure 1: The Y-axis is labeled 'Cosine Similarity' but the values for the 'RCORE' series drop to -0.79, which is inconsistent with standard cosine similarity ranges ([-1, 1]) unless specific normalization is applied, yet no explanation is provided.
- [**writing**] Figure 1: The legend labels 'C2C (Baseline)' and 'RCORE (Ours)' are undefined; the caption does not explain what these acronyms stand for or the experimental setup.
- [**writing**] Figure 2: The figure has no caption text (only the filename), making it impossible to understand the context, dataset, or specific metrics being plotted.
- [**writing**] Figure 2: The legend is placed inside the plot area, obscuring the data points and trend lines for the 'False Co-occurrence Prediction' and 'Unseen Accuracy' series.
- [**science**] Figure 2: The left y-axis (Accuracy) and right y-axis (False Prediction Ratio) share the same numerical scale range (0-50 vs 30-70 roughly), but the visual alignment of the ticks is ambiguous, potentially misleading the reader about the relative magnitude of the 'False Seen Prediction' vs 'Seen Accuracy'.
- [**writing**] Figure 3: The figure lacks a descriptive caption; it is labeled '(no caption)' and only provides a filename, making the specific context of the 'diagnosis' unclear.
- [**writing**] Figure 3: The legend entry 'False Co-occurrence Prediction' is not explicitly defined in the caption, leaving the distinction between 'False Seen' and 'False Co-occurrence' ambiguous to the reader.

- **[writing]** Figure 4: The provided caption is empty ('no caption'), making it impossible to verify if the legends, symbols, or data series (e.g., 'Verb @ Seen Comp.', 'Random Accuracy') match the authors' intended description.
- **[writing]** Figure 4: The heatmap legends (e.g., 'Present/Absent', 'Seen/Unseen') are placed directly above the plots without clear visual separation or titles, which may confuse readers regarding which legend applies to which specific grid.
- **[writing]** Figure 5: The caption is missing; the provided text '(no caption)' prevents verification of the figure's content, context, or the meaning of the visual elements.
- **[science]** Figure 5: The image displays a video frame sequence with a time axis and a text label '(Take, Cup)', but without a caption, it is unclear what specific scientific claim or data this visual evidence supports.
- **[writing]** Figure 6: The x-axis labels are rotated but still overlap significantly, making the text difficult to read; consider increasing spacing or reducing the number of labels.
- **[writing]** Figure 6: The caption is empty ('no caption'), providing no context for the data shown or its relevance to the paper's claims.
- **[writing]** Figure 7: The caption is missing; the provided text '(no caption)' does not describe the plot's content, methods, or significance.
- **[writing]** Figure 7: The legend uses undefined acronyms ('C2C', 'RCORE', 'FSP', 'FCP') without explanation in the caption or figure text.
- **[writing]** Figure 8: The caption is empty ('no caption'), providing no explanation for the complex architecture diagrams, variable definitions (e.g., L_{TORC}, L_{CPR}), or the relationship between the three panels.
- **[writing]** Figure 8: The text '1 - lambda' in panel (b) is rendered in a standard serif font, while the surrounding mathematical notation uses a sans-serif font, creating visual inconsistency.
- **[writing]** Figure 9: The figure lacks a descriptive caption; the provided metadata '(no caption)' fails to explain the 'Training Data', 'Closed-world Evaluation', and 'Open-world Evaluation' panels or the specific meaning of the hatched, solid, and 'X' marked cells.
- **[writing]** Figure 9: The legend for the 'Training Data' panel is incomplete, showing only 'Train Pairs' while omitting definitions for the empty white cells which represent unobserved pairs.
- **[writing]** Figure 11: The caption is missing; the figure contains two complex heatmaps defining 'Training Set' vs 'Test Set' splits but lacks any text description to explain the experimental setup or the significance of the 'Present/Absent' and 'Seen/Unseen' labels.
- **[science]** Figure 11: The 'Test Set' heatmap uses a 'Seen' (orange) vs 'Unseen' (blue) legend, but the diagonal cells are colored 'Seen' while the off-diagonals are 'Unseen'. This implies a specific compositional split (e.g., seen actions on seen objects), but without a caption, the

exact definition of the 'Unseen' condition (e.g., unseen action-object pairs vs. unseen objects) is ambiguous.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-structured and the technical content is accessible to a researcher in a neighboring field (e.g., video understanding or general machine learning). However, there are several instances where acronyms and dataset shorthand are used before being defined, which creates unnecessary friction for an adjacent-field reader.

Specifically, the acronyms ZS-CAR, FSP, FCP, CPR, and TORC appear in the Abstract or Introduction before their full names are provided. While the full names are eventually given in the body text, the standard convention is to define an acronym at its very first use. For example, "ZS-CAR" appears in the first sentence of the Abstract without expansion. Similarly, "FSP" and "FCP" are introduced in the Introduction as if they are standard terms, but they are specific diagnostic metrics defined later in Section 3.2.

The dataset names "Sth-com" and "EK100-com" are also used frequently without initial expansion. While "Something-Something" and "EPIC-KITCHENS" are well-known, the specific "-com" suffix denotes a specific compositional split/benchmark created or repurposed by the authors, which is not immediately obvious to an outsider.

Finally, the symbol Δ_{CG} (referred to as Dcg in some text) is used in equations and text. While the concept of a "gap" is intuitive, the specific notation should be explicitly defined alongside the term "Compositional Gap" when first introduced.

Addressing these points by ensuring every acronym and specialized symbol is defined at its first occurrence will make the paper significantly more self-contained and accessible.

Action items.

- [**writing**] Abstract & Section 1: The acronym 'ZS-CAR' is used before being defined. Define 'Zero-Shot Compositional Action Recognition (ZS-CAR)' at its first occurrence in the Abstract or Introduction.
- [**writing**] Section 1 & 3: The acronyms 'FSP' and 'FCP' are used in the Introduction and Section 3 without prior expansion. Define 'False Seen Prediction (FSP)' and 'False Co-occurrence Prediction (FCP)' at their first occurrence in Section 3.2.
- [**writing**] Section 1 & 4: The acronyms 'CPR' and 'TORC' are introduced in the Introduction and used in Section 4 without expansion. Define 'Co-occurrence Prior Regularization (CPR)' and 'Temporal Order Regularization for Composition (TORC)' at their first occurrence in Section 4.
- [**writing**] Section 3.2: The symbol 'Dcg' (or Δ_{extCG}) is used in the text and equations. Ensure

'Compositional Gap (Δ_{extCG})' is explicitly defined with the symbol before Equation 2.

- [**writing**] Section 5.1: The abbreviation 'H.M.' is used in the text and tables without definition. Define 'Harmonic Mean (H.M.)' at its first occurrence in Section 5.1.
- [**writing**] Section 5.1: The dataset names 'Sth-com' and 'EK100-com' are used as shorthand. Define 'Sth-com (Something-Something V2 compositional benchmark)' and 'EK100-com (EPIC-KITCHENS-100 compositional benchmark)' at their first occurrence.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a logically coherent argument structure. The diagnosis of "object-driven shortcuts" in Section 3 is consistently linked to the proposed solution components (CPR and TORC) in Section 4, and the experimental results in Section 5 directly address the specific failure modes identified earlier.

Specifically, the definition of the "Compositional Gap" (Δ_{CG}) in Section 3.2 (Eq. 2) is applied consistently in the results tables (e.g., Table 1, Table 3) and the ablation studies. The logic that a negative Δ_{CG} indicates a failure to model compositionality beyond independent parts is maintained throughout the text.

The causal claims regarding the proposed method's efficacy are supported by the ablation studies (Table 4), which isolate the contributions of CPR and TORC. The text correctly interprets the ablation data: for instance, the claim that TORC yields the largest gain in verb generalization aligns with the data in Table 4(b), where removing TORC causes a significant drop in `verb@unseen-comp`.

There are no contradictions between the abstract, introduction, and conclusion regarding the scope of the results. The distinction between "seen" and "unseen" compositions is rigorously maintained in definitions, metrics, and result reporting. The argument that the open-world evaluation protocol is necessary to expose shortcut behaviors is logically sound and consistently applied in the comparison with baselines.

No logical gaps, non-sequiturs, or internal contradictions were found. The reasoning holds together as a valid argument.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper presents a strong diagnostic framework for object-driven shortcuts in Zero-Shot Compositional Action Recognition (ZS-CAR) and proposes a method (RCORE) that empirically reduces these shortcuts on two specific benchmarks. However, the rhetoric occasionally exceeds the scope of the demonstrated evidence, particularly in the strength of the claims made in the contributions list and conclusion.

First, the contributions section (Introduction) claims the authors "prove that our approach mitigates co-occurrence biases." The evidence provided in Section 5 consists of empirical metrics (FSP/FCP) showing a reduction in shortcut reliance on Sth-com and EK100-com. In the context of machine learning, "proof" is a mathematical term reserved for theoretical guarantees, whereas the paper offers empirical validation. This is a classic overreach of language strength; the claim should be softened to "demonstrate" or "show" to accurately reflect the nature of the experimental results.

Second, the Abstract and Conclusion generalize the findings to "across datasets" and "across backbones" without sufficient qualification. The experimental section (Section 5) validates the method exclusively on Sth-com and the newly introduced EK100-com, using CLIP-B/16 and InternVideo2-B/14 (and one 1B variant). While these are significant benchmarks, the phrasing implies a universality that extends to any dataset or model architecture, which is not supported by the data. The claim should be scoped to the specific datasets and model families tested, or the authors should acknowledge the limitation that generalization to other domains (e.g., egocentric vs. third-person, or other video foundation models) remains untested.

Finally, the Conclusion frames the work as moving the field "toward more reliable verb–object compositional reasoning." While the method improves performance, the paper's own failure analysis (Appendix, Tables on failure modes) reveals that significant errors persist (e.g., high confusion between opposite verbs like "opening" and "closing" on unseen compositions). The conclusion should be more nuanced, acknowledging that while the proposed method mitigates specific shortcuts, the broader challenge of reliable reasoning in sparse compositional settings is not fully resolved. This would provide a more honest assessment of the current state of the art.

Action items.

- [**writing**] The paper presents a strong diagnostic framework for object-driven shortcuts in Zero-Shot Compositional Action Recognition (ZS-CAR) and proposes a method (RCORE) that empirically reduces these shortcuts on two specific benchmarks. However, the rhetoric occasionally exceeds the scope of the demonstrated evidence, particularly in the strength of the claims made in the contributions list and conclusion. First, the contributions section (Introduction) claims the authors "prove that our approach miti

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper presents a methodological contribution to computer vision (Zero-Shot Compositional Action Recognition) focused on mitigating "object-driven shortcuts" in model training. The work involves training models on existing public video datasets (Something-Something V2, EPIC-KITCHENS-100) and introducing a new benchmark split (EK100-com) derived from these public sources.

From a safety and ethics perspective, the research is low-risk:

1. **Data Provenance:** The datasets used (Sth-com, EK100) are derived from public benchmarks with established licenses. The paper does not scrape new data from social media or private sources, nor does it release any raw video data containing Personally Identifiable Information (PII). The "EK100-com" is a re-splitting of existing public data, not a new collection of human subjects.
2. **Dual-Use:** The proposed method (RCORE) aims to improve the temporal reasoning of action recognition models. While improved action recognition could theoretically be used in surveillance, the paper does not introduce capabilities that meaningfully lower the barrier to harmful surveillance or deception compared to existing state-of-the-art models. The method is a regularization technique for training, not a tool for generating deceptive content or exploiting vulnerabilities.
3. **Human Subjects:** The research does not involve direct interaction with human subjects, surveys, or the collection of private behavioral data. The use of public video datasets for algorithmic training does not require IRB approval in this context, and no such statement is missing.
4. **Bias/Fairness:** The paper explicitly addresses a form of bias (co-occurrence priors leading to shortcut learning) and proposes a method to mitigate it. It does not introduce new biases against identifiable demographic groups.

There are no missing disclosures, unmitigated risks, or ethical violations specific to this work. The paper is a standard algorithmic improvement study with no foreseeable non-trivial safety risks that require mitigation or disclosure beyond what is standard for the field.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling diagnosis of object-driven shortcuts in Zero-Shot Compositional Action Recognition (ZS-CAR) and proposes a method (RCORE) to mitigate them. The experimental design is generally sound, utilizing two datasets (Sth-com, EK100-com) and multiple backbones. However, the evidentiary strength of the reported improvements is weakened by a lack of statistical robustness and incomplete isolation of the proposed components' effects.

First, the primary results in Tables 1 and 2 report single-point accuracy improvements (e.g., +3.8% on unseen compositions for Sth-com with CLIP) without any indication of variance. In deep learning benchmarks, especially with complex training objectives like RCORE, performance can fluctuate significantly across random seeds. A 3-4 point gain is meaningful but could plausibly arise from a lucky initialization or a specific random seed if not averaged over multiple runs. The paper should report results as mean ± standard deviation (or standard error) across at least 3-5 independent training runs with different seeds. Without this, the reader cannot distinguish a robust methodological improvement from statistical noise.

Second, the ablation studies, while detailed, do not fully isolate the contribution of the data augmentation strategy in Co-occurrence Prior Regularization (CPR) from the regularization loss itself. Section 4.1 describes synthesizing new videos by mixing frames (Eq. 1). Table 3(c) compares the full method against a "Full Open-world" baseline and a "Closed-world" baseline, but it does not include a control run that applies the *same* frame-mixing augmentation to the baseline model *without* the specific CPR margin loss (L_{CPR}). It is possible that the performance gain attributed to CPR comes simply from the regularization effect of training on augmented data (a form of Mixup/CutMix) rather than the specific mechanism of penalizing frequent co-occurrences. To claim that the *specific* CPR design is responsible, the authors must show that the augmentation alone (without the loss) yields less improvement than the full CPR method.

Finally, the Temporal Order Regularization (TORC) relies on specific hyperparameters, notably the temperature τ in the entropy loss and the specific permutation strategy for shuffling. The ablation in Table 3(b) isolates the loss terms (L_{cos} vs L_{ent}) but does not explore the sensitivity to τ or the nature of the shuffling. If the gain is highly sensitive to a specific τ value or a specific shuffling distribution, the robustness of the claim is reduced. A brief sensitivity analysis or a justification for the chosen τ would strengthen the evidence that the method is not over-tuned to a specific configuration.

Addressing these points—specifically adding seed variance and a control for the CPR augmentation—would significantly bolster the claim that RCORE's improvements are due to the proposed mechanisms rather than experimental artifacts or luck.

Action items.

- **[writing]** The paper presents a compelling diagnosis of object-driven shortcuts in Zero-Shot Compositional Action Recognition (ZS-CAR) and proposes a method (RCORE) to mitigate them. The experimental design is generally sound, utilizing two datasets (Sth-com, EK100-com) and multiple backbones. However, the evidentiary strength of the reported improvements is weakened by a lack of statistical robustness and incomplete isolation of the proposed components' effects. First, the primary results in Tables 1 and

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in this paper lacks the rigor required to substantiate claims of "significant" improvement or to distinguish signal from noise.

First, the paper reports single point estimates for accuracy, FSP, FCP, and Compositional Gap in Section 5 and Tables 1-4 without any measure of uncertainty (standard deviation, standard error, or confidence intervals). In deep learning, where stochasticity can lead to variance of 1-3% or more, a 4-point gap (e.g., 34% vs 30%) from a single run is not statistically distinguishable from random fluctuation. The authors must report results as mean ± standard deviation over at least 3-5 independent seeds to validate the stability of the proposed method.

Second, the term "significantly" is used repeatedly (e.g., "significantly improves unseen composition accuracy") without supporting statistical evidence. No hypothesis tests (e.g., paired t-test, Wilcoxon) or p-values are reported. Without a formal test and a stated alpha level, this terminology is misleading. The authors should either perform these tests and report the results or rephrase their claims to describe the magnitude of the improvement without invoking statistical significance.

Finally, the ablation studies in Section 5.3 involve multiple comparisons across different loss components and hyperparameters. The paper highlights a "best" configuration without addressing the multiple testing problem, which increases the risk of false positives. The authors should apply a correction (e.g., Bonferroni) or explicitly acknowledge this limitation when interpreting the results.

Action items.

- [**writing**] Section 5 and Tables 1-4 report single point estimates (e.g., '34% vs 30%') without uncertainty measures (SD/SE/CI) across seeds. Deep learning results require variance reporting to distinguish signal from noise. Report mean ± SD over 3 seeds or explicitly state results are from a single run.
- [**writing**] The paper claims 'significantly improves' (e.g., Sec 5.2) without reporting p-values or hypothesis tests. Statistical significance requires a formal test (e.g., paired t-test) and alpha level. Either run the test and report p-values or rephrase claims to describe magnitude (e.g., 'improves by X points') without invoking significance.
- [**writing**] Ablation studies in Sec 5.3 compare multiple configurations without correcting for multiple comparisons. Highlighting the 'best' configuration among many tests inflates false-positive risk. Apply a correction (e.g., Bonferroni) or explicitly acknowledge the uncorrected nature of these comparisons.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured and the argument flows logically from the diagnosis of the problem to the proposed solution. However, there are several instances where sentence construction impedes immediate comprehension, forcing the reader to re-parse or guess the intended structure.

In Section 3.2, the description of the bias-controlled experiment is buried in a long, complex sentence that delays the main action. Similarly, in Section 4, the introduction of the backbone adapters is muddled by awkward phrasing ("employ an adapter-based tuning") and a cluttered list structure. These are not scientific errors but clear writing defects that slow down the reader.

Additionally, there are minor issues with consistency and clarity. A typo in the backbone name ("InternVi-deo2") in Section 5.1 is distracting. In Section 5.2, the explanation of the temporal subset evaluation is unnecessarily verbose, repeating the goal of the experiment within the same sentence. Finally, in the Appendix, a logical jump in the EK100-com description leaves the reader wondering how single verb-object pairs lead to "explicit object supervision" without further explanation.

Addressing these specific sentence-level issues will significantly improve the readability and professional polish of the manuscript.

Action items.

- [**writing**] The paper is generally well-structured and the argument flows logically from the diagnosis of the problem to the proposed solution. However, there are several instances where sentence construction impedes immediate comprehension, forcing the reader to re-parse or guess the intended structure. In Section 3.2, the description of the bias-controlled experiment is buried in a long, complex sentence that delays the main action. Similarly, in Section 4, the introduction of the backbone adapters is mud